

ATPL EASA STUDY GUIDE | 050 - METEOROLOGY

Chapter 5: Wind Dynamics, Coriolis & Local Wind Systems

Syllabus Source: CAE Oxford (pp. 137-174) | Pass Target: $\geq 90\%$ AviationExam

1. FORCES GOVERNING WIND & BALANCE

Air motion is governed by four fundamental forces:

- Pressure Gradient Force (PGF): Acts perpendicular to isobars from HIGH pressure towards LOW pressure. Initiates all horizontal air motion.
- Coriolis Force: Apparent force caused by Earth's rotation. Deflects moving air to the RIGHT in the Northern Hemisphere (to the LEFT in the Southern Hemisphere). Zero at equator; maximum at poles. Proportional to wind speed.
- Geostrophic Wind: Theoretical wind resulting from an exact balance between PGF and Coriolis force blowing parallel to straight, parallel isobars above friction layer.
- Surface Friction Effect (Veering vs Backing): Friction slows surface wind, reducing Coriolis force. PGF pulls air across isobars towards low pressure ($\sim 30^\circ$ over land, $\sim 10^\circ$ over sea). In Northern Hemisphere: Climbing out, wind VEERS (turns clockwise) and INCREASES; Descending to land, wind BACKS (turns counter-clockwise) and DECREASES!

2. LOCAL WIND PHENOMENA

- Sea Breeze vs Land Breeze: Sea breeze develops by day (cooler sea air blows onshore toward heated land). Land breeze develops at night (cooling land air blows offshore).
- Anabatic vs Katabatic Winds: Anabatic wind: Daytime upslope wind as sun heats valley walls. Katabatic wind: Nighttime downslope gravity wind as dense cold air drains into valleys (e.g. Bora, Mistral).
- The Föhn Effect (Chinook): Moist air rises on windward mountain slope cooling at SALR (losing moisture by rain). Descending on leeward slope, dry air warms at DALR ($3^\circ\text{C}/1,000\text{ ft}$), arriving warm and extremely dry!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Surface Friction Effect & Buys Ballot's Law

Rule 1 (Buys Ballot's Law - Northern Hemisphere): - Stand with your back to the wind: LOW PRESSURE is on your LEFT (slightly ahead at surface). Rule 2 (Friction Layer Wind Shift in Descent): - Descending from 3,000 ft AGL to the runway, ground friction slows the wind, reducing Coriolis. - The wind BACKS (turns counter-clockwise / to the left) by approx 30 degrees over land and speed halves. - On departure (climbing): The wind VEERS (turns clockwise / to the right) and doubles in speed.

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Calculating Temperature & Dew Point in a Föhn Wind

SCENARIO (Standard EASA Föhn Exam Problem): Air at sea level (0 ft) on the windward side of a 6,000 ft mountain has: - Surface Temperature = +18°C - Surface Dew Point = +10.5°C - Cloud base (condensation level) on windward slope = 3,000 ft - Air crosses the ridge at 6,000 ft and descends to sea level on the leeward side QUESTION: What will be the temperature and dew point of the air when it reaches sea level on the leeward side? SOLUTION PASO A PASO (FOR DUMMIES): Part 1: Windward Ascent (0 to 6,000 ft): - From 0 to 3,000 ft (Dry ascent below cloud base at DALR = 3°C/1,000 ft): * Temp decrease = $3 \times 3.0^\circ\text{C} = 9.0^\circ\text{C}$. * Temp at 3,000 ft = $18.0^\circ\text{C} - 9.0^\circ\text{C} = +9.0^\circ\text{C}$. - From 3,000 to 6,000 ft (Saturated ascent inside cloud at SALR = 1.5°C/1,000 ft): * Temp decrease = $3 \times 1.5^\circ\text{C} = 4.5^\circ\text{C}$. * Temp at mountain top (6,000 ft) = $9.0^\circ\text{C} - 4.5^\circ\text{C} = +4.5^\circ\text{C}$. * (Because it is saturated at the peak, Dew Point at 6,000 ft = +4.5°C!). Part 2: Leeward Descent (6,000 ft down to 0 ft): - The air dropped its moisture as rain on the windward side! All clouds evaporate instantly upon descending. - ENTIRE DESCENT IS DRY at DALR = 3.0°C / 1,000 ft! - Temperature increase = $6 \times 3.0^\circ\text{C} = +18.0^\circ\text{C}$. - Final Leeward Temp at sea level = $+4.5^\circ\text{C} + 18.0^\circ\text{C} = +22.5^\circ\text{C}$! Part 3: Final Dew Point on leeward side: - Dew point increases down the slope at 0.5°C / 1,000 ft = $6 \times 0.5^\circ\text{C} = 3.0^\circ\text{C}$. - Final Leeward Dew Point = $4.5^\circ\text{C} + 3.0^\circ\text{C} = +7.5^\circ\text{C}$. FINAL ANSWER: Leeward surface air is dramatically WARMER (+22.5°C vs +18°C) and DRIER (Dew Point +7.5°C vs +10.5°C)! Classic Föhn wind.

[!] EXAM TRAP (AVIATIONEXAM) - Veering vs Backing Memory Trap

- In the Northern Hemisphere: - VEER = Clockwise turn (e.g. 240° to 270°). Occurs when CLIMBING. - BACK = Counter-clockwise turn (e.g. 270° to 240°). Occurs when DESCENDING towards landing.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: In the Northern Hemisphere, what happens to the wind direction and speed when descending from 3,000 ft

- [A]: Veers and increases
- [B]: BACKS (turns counter-clockwise) and DECREASES in speed due to surface friction
- [C]: Veers and decreases
- [D]: Remains unchanged

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Surface friction slows the wind, reducing the Coriolis force. The pressure gradient force pulls the wind towards lower pressure, causing it to back and reduce speed.

Q2: What is the primary characteristic of a KATABATIC wind?

- [A]: Warm daytime wind blowing upslope
- [B]: COLD, DENSE AIR flowing downhill at night under the force of gravity
- [C]: Wind blowing from sea to land
- [D]: Jet stream core

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Nocturnal terrestrial radiation cools mountain air, making it dense and heavy. Gravity pulls this cold pool down valleys and slopes as a katabatic wind.

Q3: In the Föhn effect, why is the air arriving on the leeward side of the mountain warmer than when it started?

- [A]: Ground friction heats the air
- [B]: Air cools at SALR on the wet ascent (releasing latent heat), but warms at the STEEPER DALR during the entire dry descent
- [C]: Solar radiation doubles
- [D]: Coriolis force

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Latent heat released during cloud formation on ascent warms the parcel; during descent, cloud dissipation allows dry adiabatic warming (3°C/1000 ft) over the full height.